

87 $\lambda(\cap A_n) = 1$ (claim)

$\lambda: \sigma \subseteq \mathcal{P}(X) \rightarrow \mathbb{R}^+ \cup \{0\}$

$(\cap A_n)^c = \cup A_n^c$

$\lambda(A_n) = 1$

$\lambda(A_n^c)$

$= 1 - \lambda(A_n)$
 $= 1 - 1$
 $= 0$

$\Rightarrow \lambda((\cap A_n)^c) = \lambda(\cup A_n^c)$
 $0 \leq \leq \sum \lambda(A_n^c)$

$\Rightarrow \lambda((\cap A_n)^c) = 0 \leq 0$

$\Rightarrow 1 - \lambda(\cap A_n) = 0$

$\Rightarrow \boxed{\lambda(\cap A_n) = 1}$

If possible let $\cap A_n$ is not dense in $[0,1]$.

Then $\exists x \in [0,1], x \notin \overline{\cap A_n}$.

$\exists \varepsilon > 0$ s.t. $(x - \varepsilon, x + \varepsilon) \cap \cap A_n = \emptyset$.

$\Rightarrow (x - \varepsilon, x + \varepsilon) \subseteq (\cap A_n)^c$

$\Rightarrow \lambda(x - \varepsilon, x + \varepsilon) \leq \lambda(\cap A_n^c)$

$\Rightarrow 2\varepsilon \leq 0 \Rightarrow \varepsilon \leq 0 \Rightarrow \varepsilon = 0$

$$\Rightarrow \overline{\bigcap A_n} = [0, 1].$$